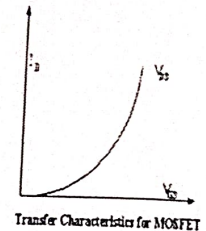
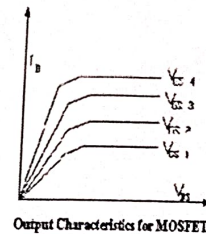
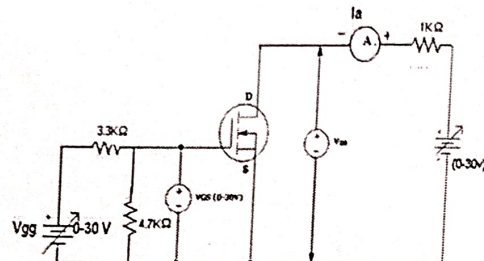


Circuit diagram for MOSFET:

EXPERIMENT 1

CIRCUIT DIAGRAM (MOSFET):



Tabulation

Output characteristic of MOSFET

S.NO	V _{GS} (V)	I _D (mA)
1.	1	3.5
2.	3	6.8
3.	5	9.2
4.	8	7.3

Tabulations:

Output characteristics of MOSFET

$$V_{GS} = 5V$$

S.NO	V_{DS} (V)	I_D (mA)
1	1	3.5
2	3	6.8
3	5	7.2
4	8	7.7

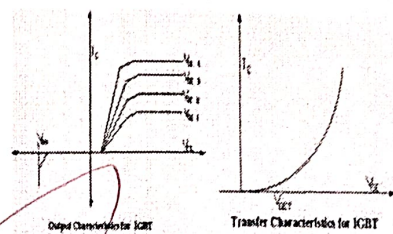
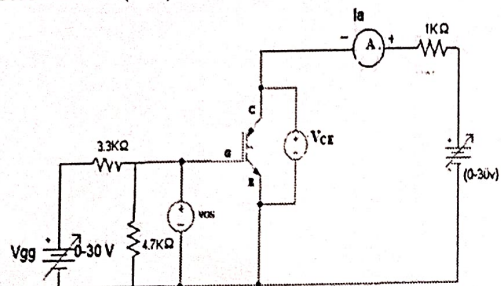
Transfer characteristics of MOSFET

$$V_{DS} = 5V$$

S.NO	V_{GS} (V)	I_D (mA)
1	2	0
2	2.5	0.8
3	3	3.5
4	3.5	7.2

EXPERIMENT 2

CIRCUIT DIAGRAM (IGBT):



Output characteristics of IGBT

$$V_{GE} = 5V$$

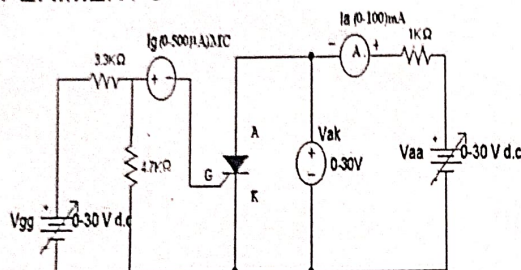
Transfer characteristics of IGBT

$$V_{GE} = 5V$$

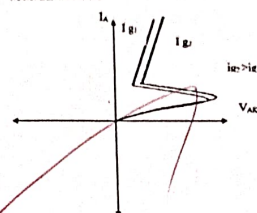
S.NO	V_{CE} (V)	I_C (mA)	S.NO	V_{CE} (V)	I_C (mA)
1	1	15	1	0	0
2	3	38	2	5	0
3	5	42	3	8	12
4	8	43	4	12	20
			5	16	50
			6	20	70
			7	25	90
			8	30	110

Circuit diagram:-

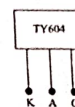
EXPERIMENT 3



MODEL GRAPH:



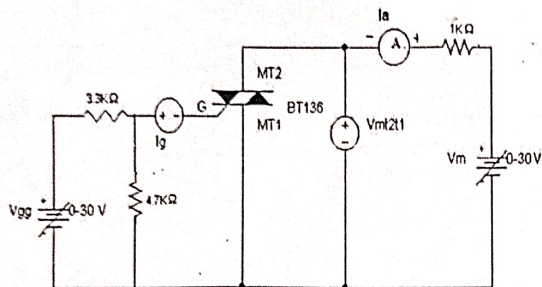
Pin configuration



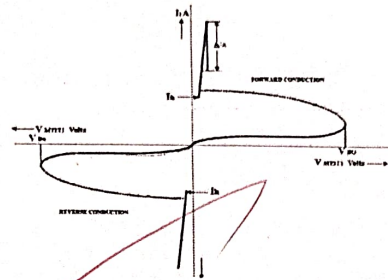
Tabular column :-

S.NO	$I_E = 6mA$		$I_E = 10mA$	
	V_{AK}	$I_A (mA)$	$V_{AK} (V)$	$I_A (mA)$
1	2	5	2	10
2	4	18	4	28
3	6	35	6	48
4	8	55	8	70
5	10	75	10	90
6	12	95	12	110
7	14	115	14	130

EXPERIMENT 4

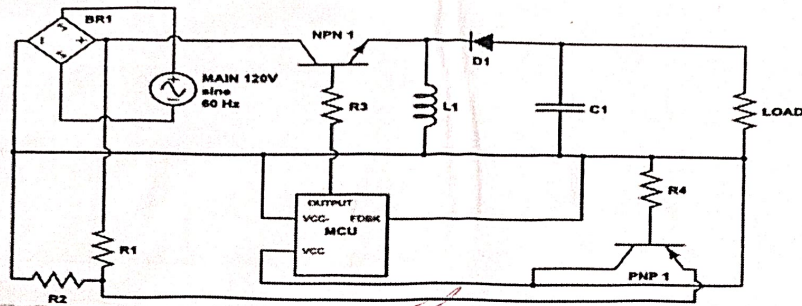


MODEL GRAPH



S.NO	$I_G = 150 \mu A$		$I_G = 150 \mu A$	
	$V_{mT_1} (V)$	$I_{mT_1} (mA)$	$V_{mT_2} (V)$	$I_{mT_2} (mA)$
1	2	10	2	8
2	4	22	4	18
3	6	38	6	30
4	8	55	8	48
5	10	70	10	65
6	12	85	12	80
7	14	100	14	95

EXPERIMENT 5



Tabulation

S. NO	T_{ON}	T_{OFF}	T	$R = T_{ON}/T$	$V_o = \frac{V_s}{1-k}$	V_o practical
1	2	8	10	0.2	2.4	2.2
2	4	6	10	0.4	4.8	4.5
3	5	5	10	0.5	6.0	5.7

Tabular column:-

$$V_s = 2.44 \text{ V}$$

S.NO	T_{ON} sec	T_{OFF} sec	T sec	Duty Ratio, $k = T_{ON}/T$	$V_o = V_s / (1-k)$. Theoretical (V)	V_o practical (V)
1	2	8	10	0.2	2.4	2.2
2	4	6	10	0.4	4.8	4.5
3	5	5	10	0.5	6.0	5.7

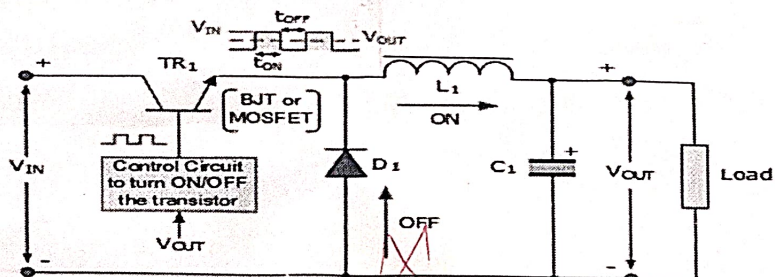
$$T = T_{ON} + T_{OFF}$$

T_{ON} - ON Time.

T_{OFF} - OFF Time.

T - Total time period.

EXPERIMENT 6



Tabular Column:-

$$V_s = \underline{11.31} \text{ V}$$

S.NO	T_{ON} sec	T_{OFF} sec	T sec	Duty Ratio, $K = T_{ON}/T$	$V_o = V_s/(1-K)$ Theoretical V	V_o practical V
1	2	8	10	0.2	15	14.2
2	4	6	10	0.4	20	18.8
3	5	5	10	0.5	24	22.5
4	6	4	10	0.6	30	28.5
5	7	3	10	0.7	40	37.5

$$T = T_{ON} + T_{OFF}$$

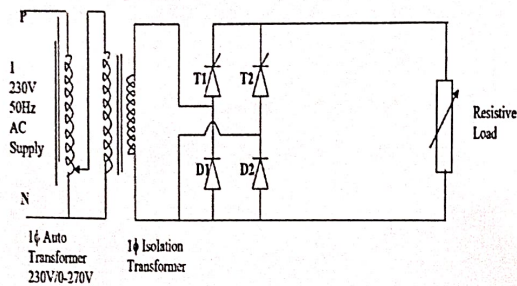
T_{ON} - On time.

T_{OFF} - Off time.

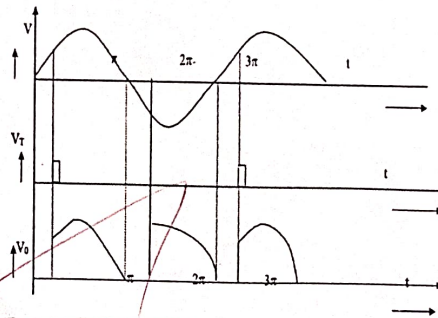
T - Total time period.

EXPERIMENT 7

CIRCUIT DIAGRAM:



MODEL GRAPH:

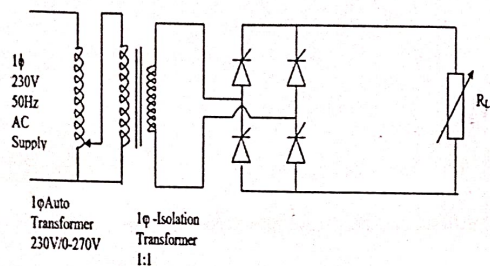


Tabulator Column:-

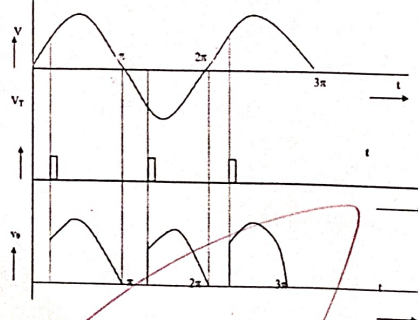
S.NO	Firing Angle Degree	Time period (T) ms		Output voltage (V)	
		T _{ON}	T _{OFF}	Practical	Theoretical
1	30°	8.33	1.67	90	96
2	60°	6.67	3.33	72	78
3	90°	5	5	48	52
4	120°	3.33	6.67	22	26
5	150°	1.67	8.33	5	7

EXPERIMENT 8

CIRCUIT DIAGRAM:



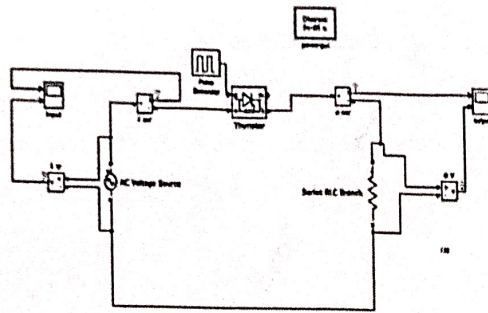
MODEL GRAPH:



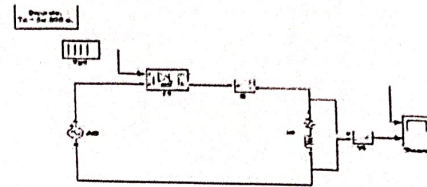
Tabular column:-

S.NO	Firing Angle Degree	Time period (T) (ms)		Output Voltage (V _o) (V)	
		T _{ON}	T _{OFF}	Practical	Theoretical
1	30°	7.5	2.5	170	170
2	60°	6	4	128	140
3	90°	5	5	95	104
4	120°	4	6	50	52
5	150°	2.5	7.5	10	12

EXPERIMENT 9

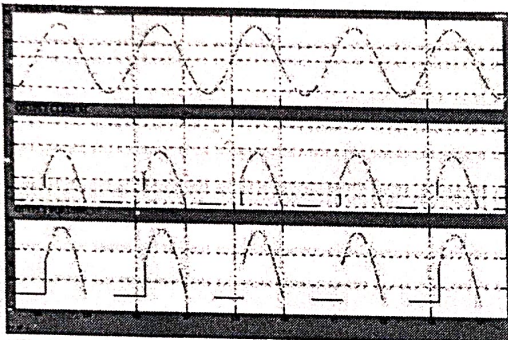


Half Wave Rectifier With RL-Load

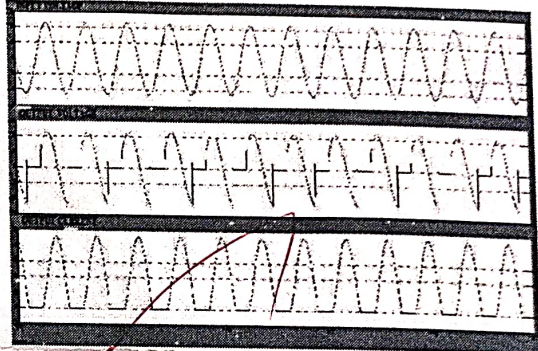


OUTPUT WAVEFORMS:

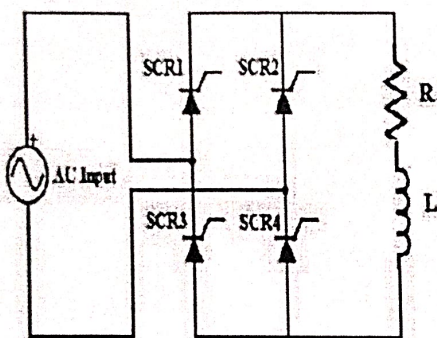
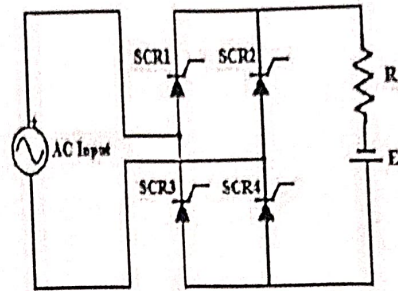
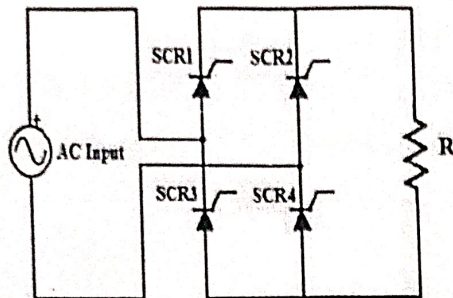
Half wave rectifier with r load:



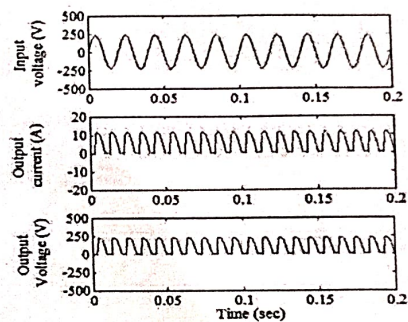
Half wave rectifier with RL load:



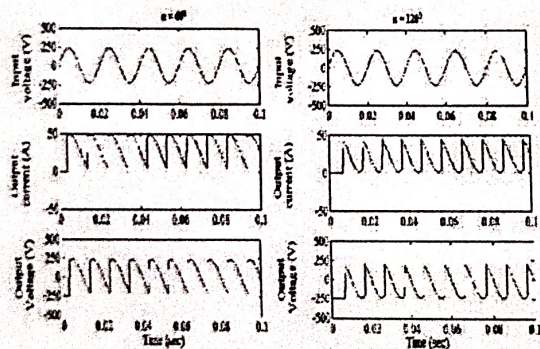
EXPERIMENT 10



OUTPUT WAVEFORMS:
Full converter fed resistive load



Full converter fed Resistive-Back Emf (RE) load at different firing angles



Full Converter fed Resistive-Inductive Load at different firing angles

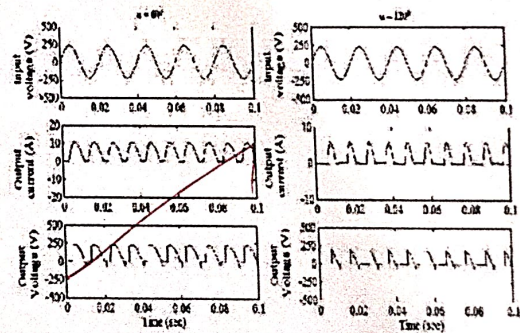


Figure 1

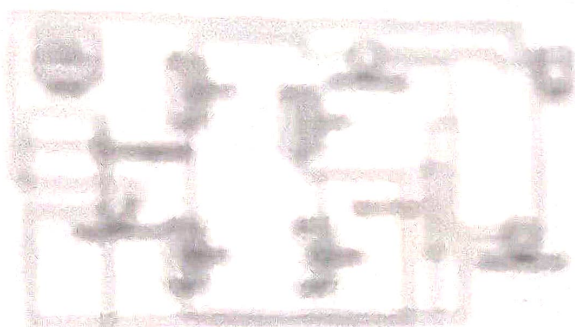
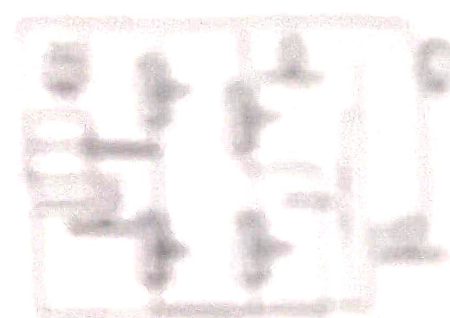


Figure 2



OUTPUT WAVEFORMS:
Full wave rectifier with r load:

